

Deliverable 3: Power Function Calculator

Code Quality, Debugging, GUI Design, Accessibility, and Unit Testing

Yichen Huang Student ID: 40167688 Function: $f(x, y) = x^y$ Version 1.1.0

Project Objective

Deliverable 3 improves the D2 implementation of the Power Function Calculator through programming-style compliance, automated code analysis, debugging, Semantic Versioning, user-interface design principles, accessibility improvements, and unit testing.

D2 Feedback Addressed

D2 Feedback	D3 Modification
Numeric conversion inside the numerical core	Moved to parse numeric input().
Valid large result rejected as overflow	Actual computed result is now validated.
Repository evidence too small	Evidence was cropped and enlarged.
Requirements table too dense	Replaced with concise evidence mapping.

Architecture

GUI → Input Parsing → Domain Validation

Numerical Core → Result / Error Feedback

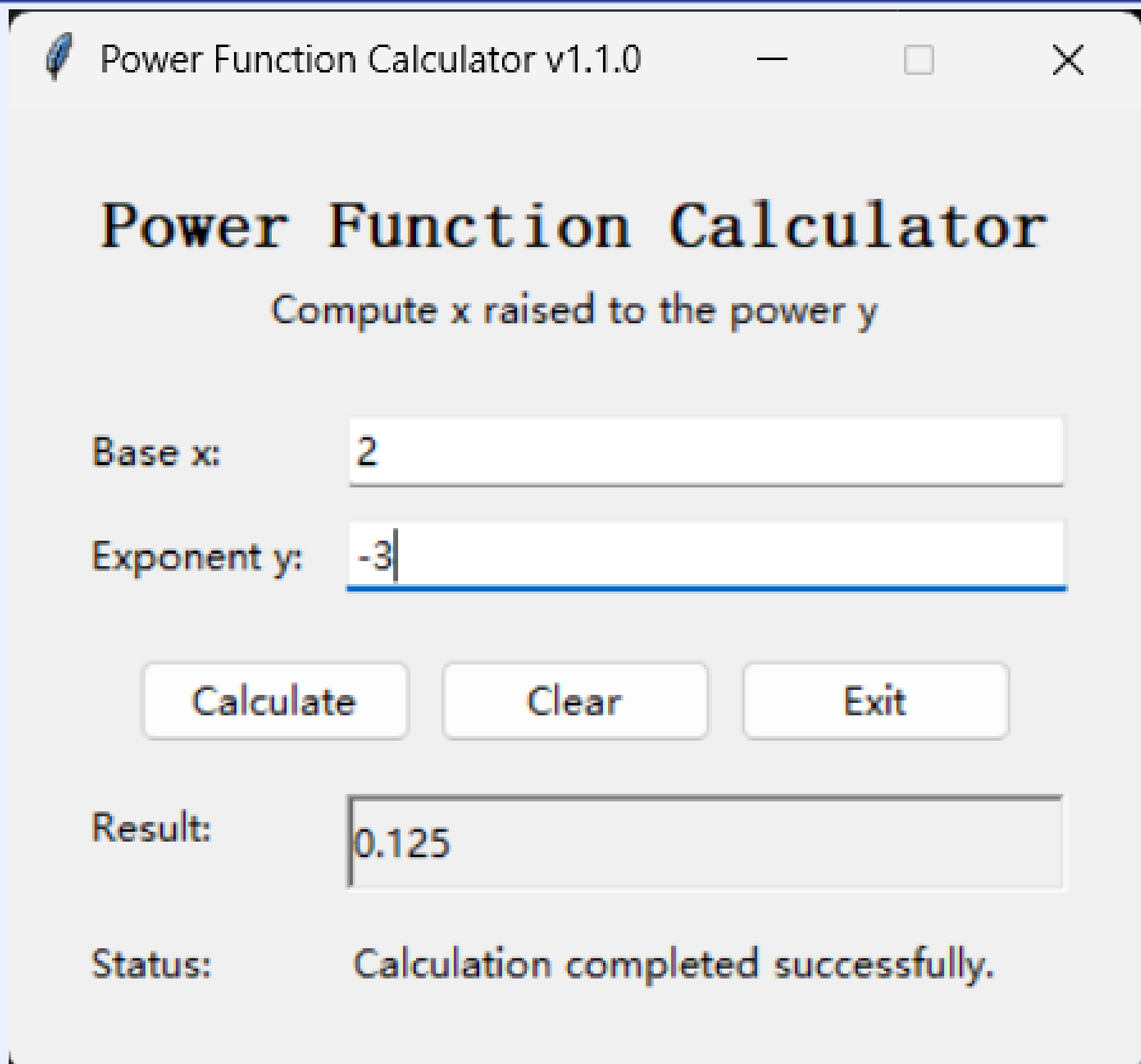
Text-to-number conversion is performed outside the numerical core. The calculation functions receive already validated numeric arguments.

Semantic Versioning

D2: 1.0.0 ⇒ D3: 1.1.0

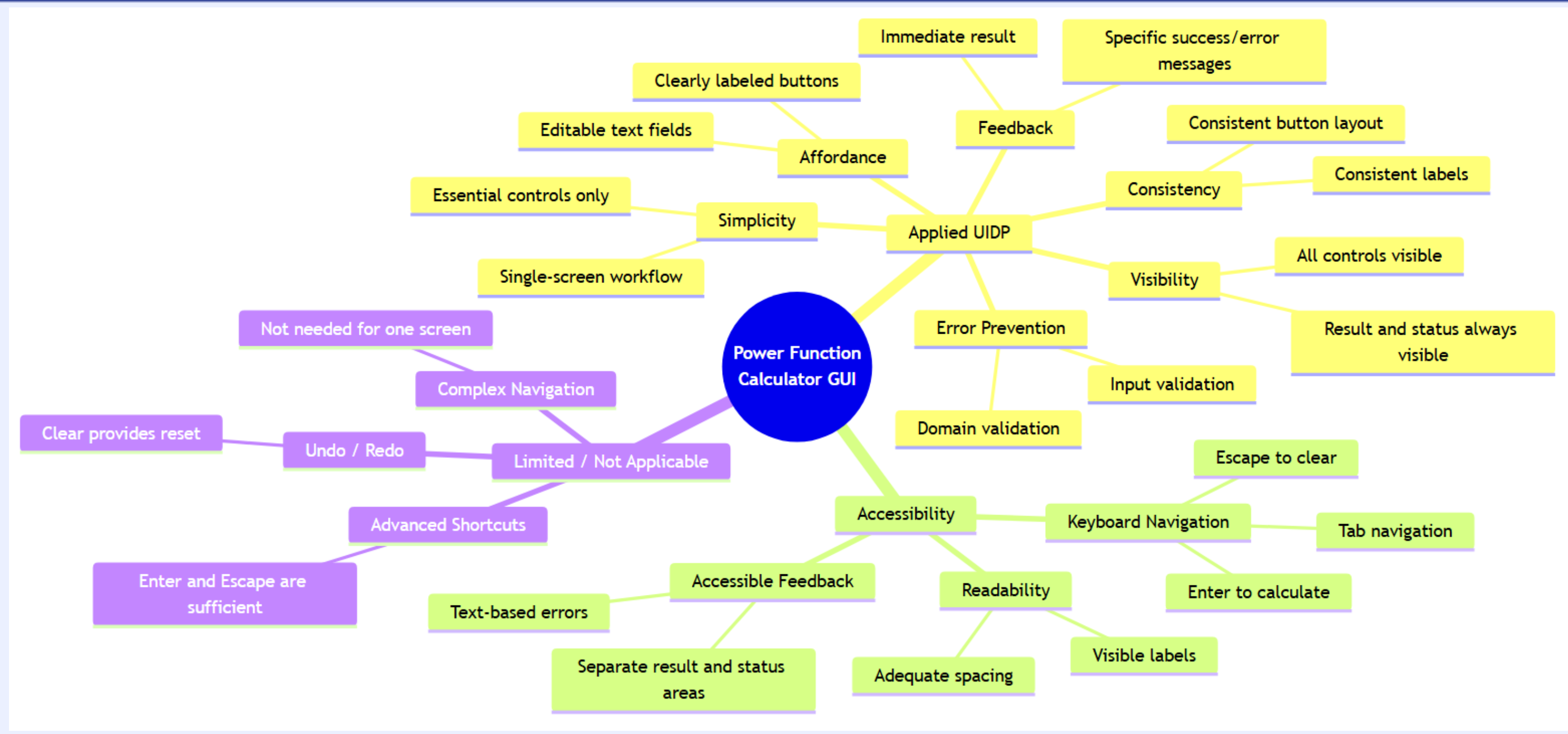
- Major version remains compatible.
- Minor version records accessibility, testing, and code-quality improvements.
- Patch remains zero for the first 1.1 release.

Final Graphical User Interface



- Clearly labelled Base, Exponent, Result, and Status fields
- Immediate success and error feedback
- Enter key performs calculation
- Escape key clears the interface
- Version 1.1.0 is visible in the window title

User Interface Design Principles



Accessibility

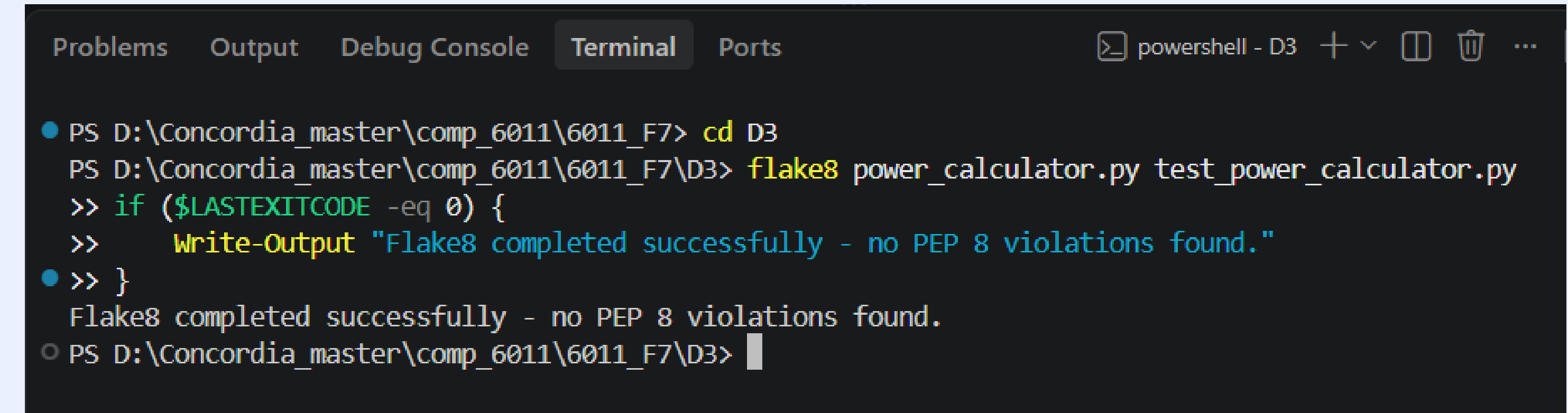
- Keyboard navigation using Tab
- Enter to calculate
- Escape to clear
- Visible keyboard focus
- Readable labels and adequate spacing
- Errors communicated through text, not colour alone
- Separate result and status areas

D3 Requirement-to-Evidence Summary

Requirement	Evidence
PEP 8 compliance	Flake8: no violations
Static analysis	Pylint: 10.00/10
Debugger usage	Negative-exponent execution path
Semantic Versioning	Version 1.1.0
UIDP and accessibility	Mind map, keyboard support, and textual feedback
Unit testing	34 PyUnit tests passed

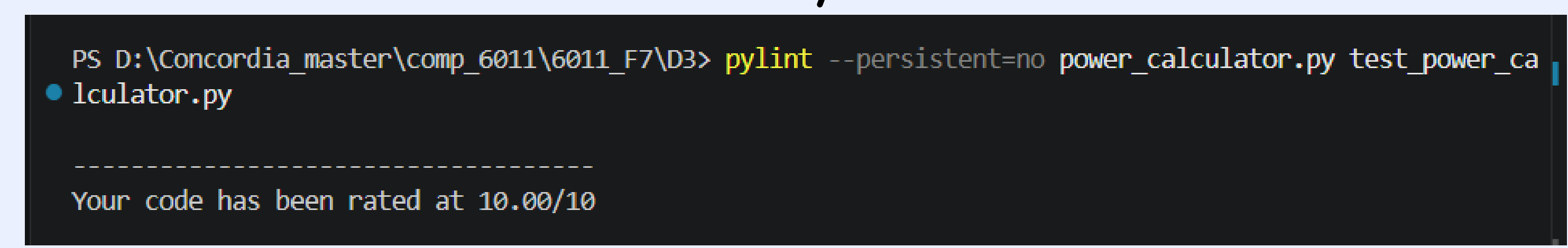
PEP 8 and Flake8

No PEP 8 Violations



Pylint Static Analysis

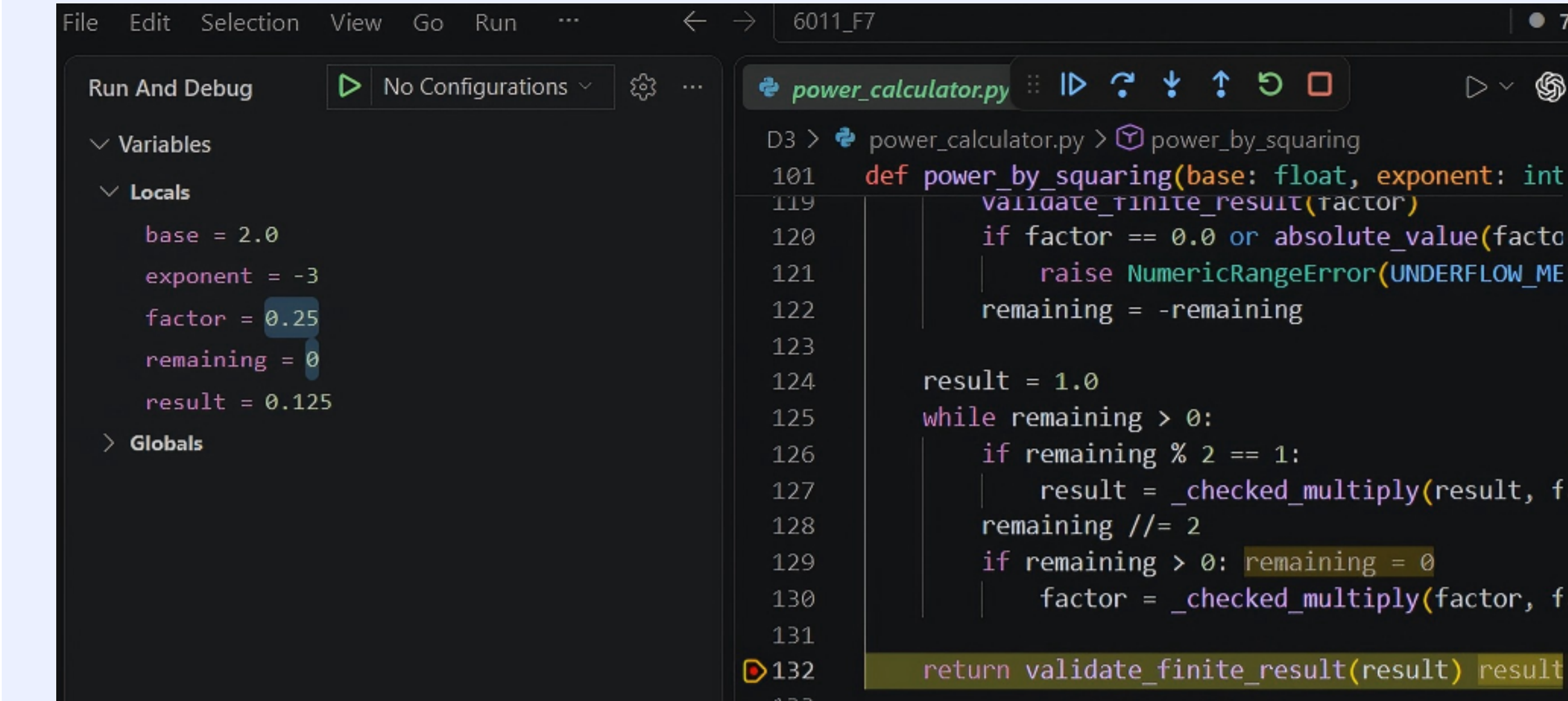
10.00 / 10



Debugger Evidence

The negative-exponent case 2^{-3} was inspected using the VS Code debugger:

- The exponent was identified as integer-valued.
- The algorithm entered the negative-exponent branch.
- The reciprocal factor was calculated.
- The final result was confirmed as 0.125.



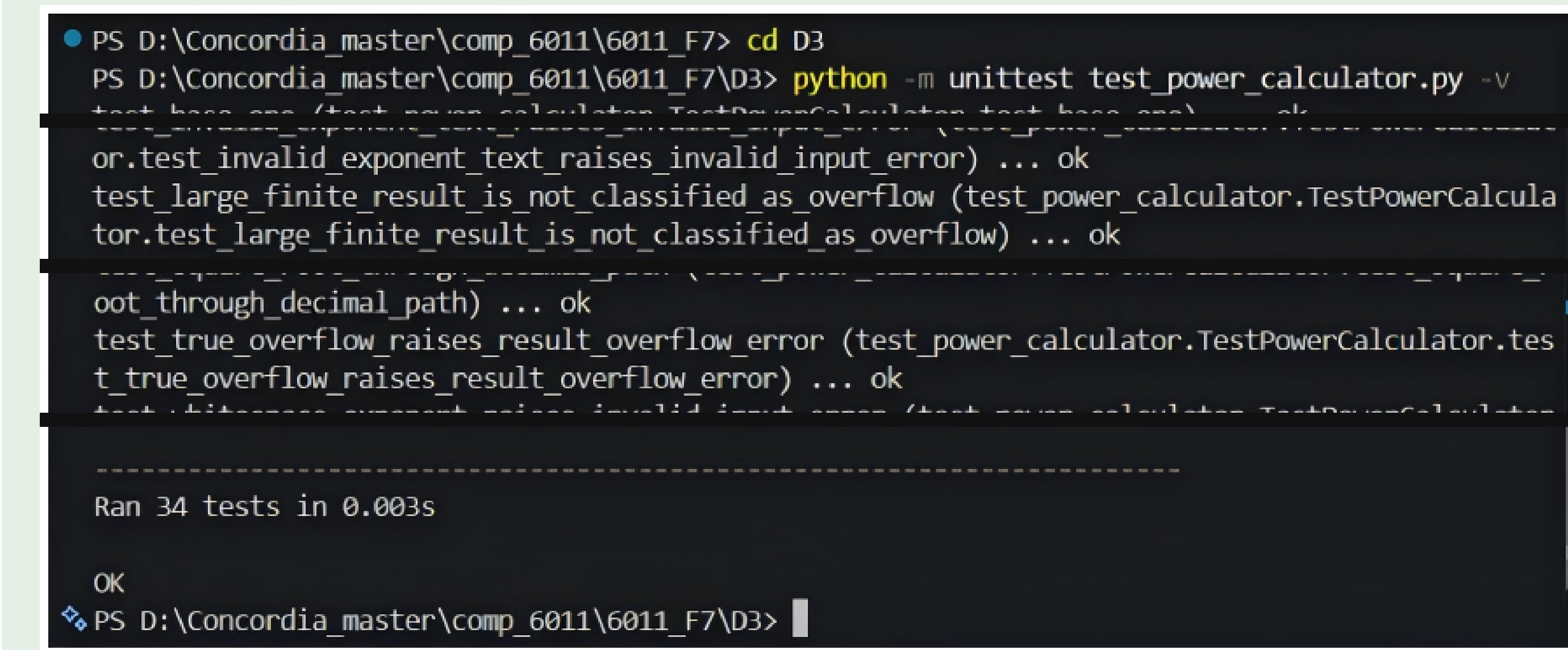
PyUnit Testing

34 Tests Passed

- Input parsing
- Integer and decimal exponents
- Unsupported domains
- NaN and infinity
- Overflow and underflow
- D2 regression tests

Regression evidence:

- Valid large finite result is accepted.
- Genuine overflow raises an error.



Conclusion

- Input conversion was separated from the numerical core.
- Overflow handling was corrected without hiding genuine errors.
- The implementation achieved zero Flake8 violations and a Pylint score of 10.00/10.
- The GUI improved keyboard access, readability, and text-based feedback.
- All 34 PyUnit tests passed, including D2 regression tests.
- Version 1.1.0 preserves the supported mathematical behaviour while improving quality and maintainability.